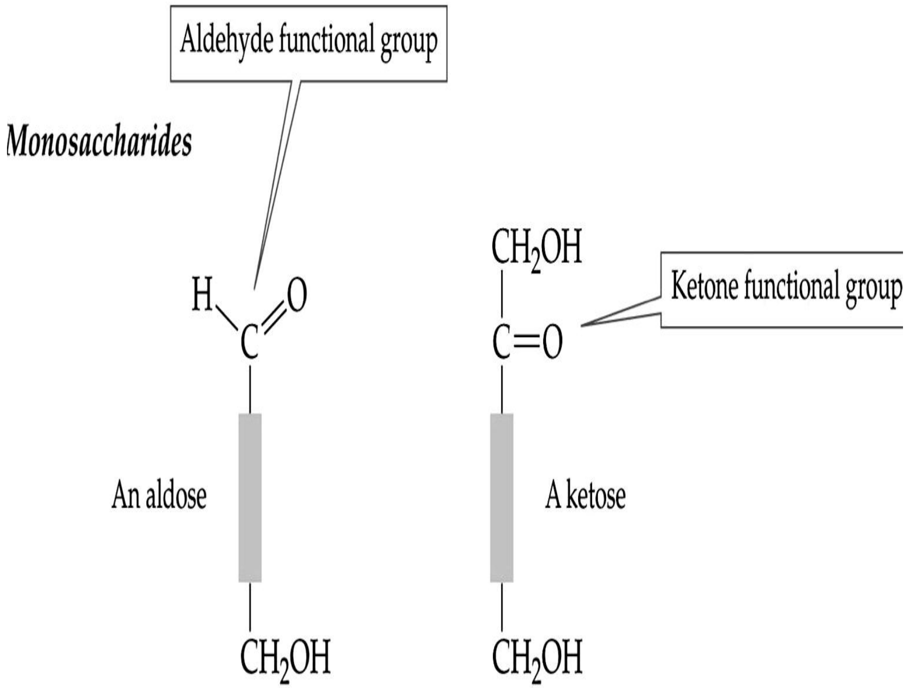
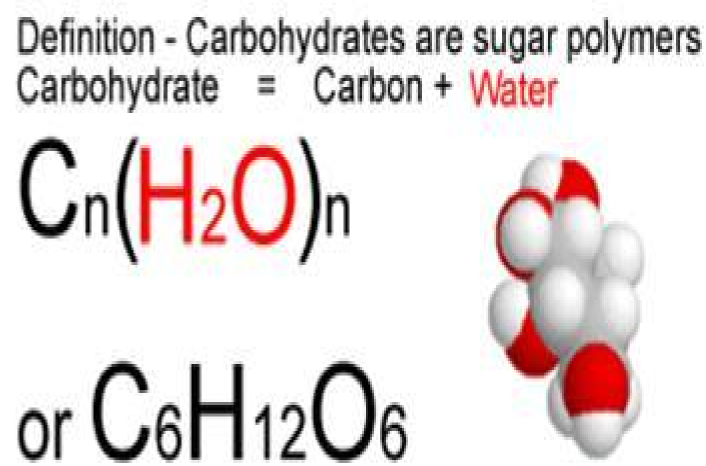
CHEMISTRY OF CARBOHYDRATES

DEFINITION

- Carbohydrates (CHO) are aldehyde or ketone derivatives of polyhydric alcohols or any substances derived from them.

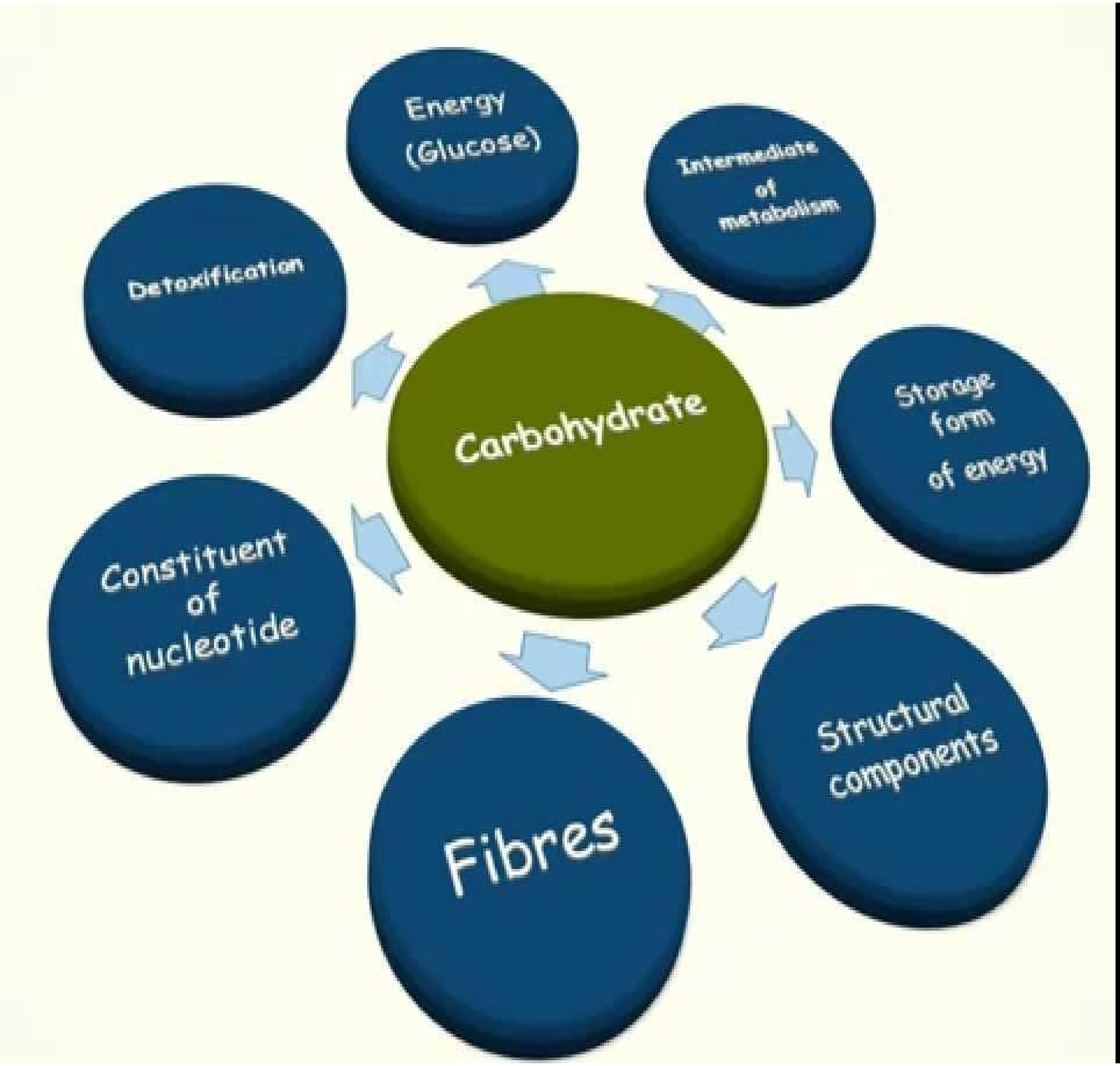




SOURCES


Simple carbohydrates
Simple carbohydrates are found in foods such as fruits, milk, and vegetables
Cake, candy, and other refined sugar products are simple sugars which also provide energy but lack vitamins, minerals, and fiber
A.D.A.M.

BIOMEDICAL IMPORTANCE

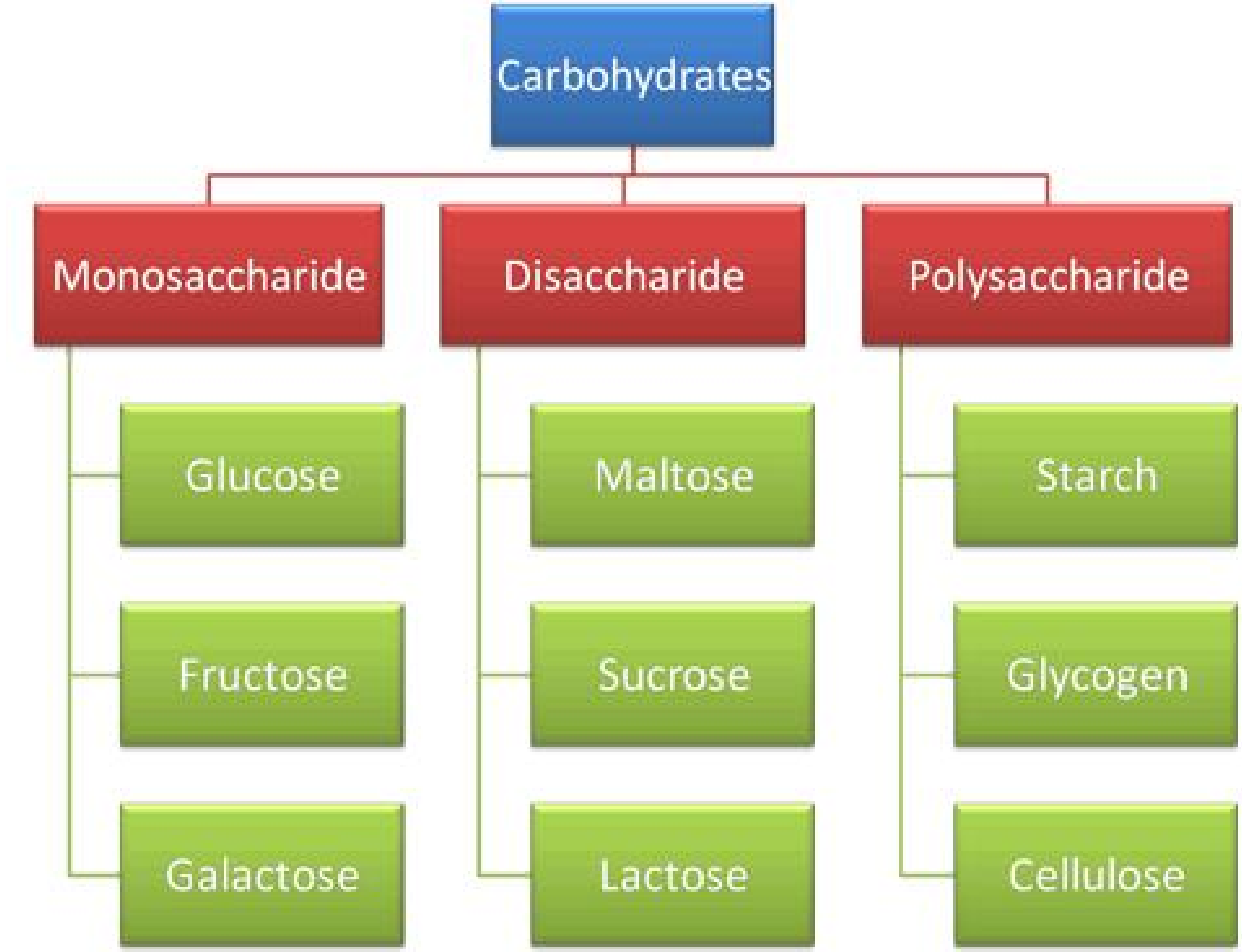


BIOMEDICAL IMPORTANCE

1. Most abundant dietary source of energy.
2. Also serve as storage form of energy –Glycogen.
3. Participate in the structure of cell membrane & cellular functions (cell growth, adhesion and fertilization).
4. Pentoses are important as part of DNA, RNA and several coenzymes.
5. Cellulose forms the cell walls of plants and is important in diet, preventing constipation.
6. Mucopolysaccharides form the ground substance of mesenchymal tissues.
7. Certain carbohydrate derivatives are used as drugs, like cardiac glycosides / antibiotics.

CLASSIFICATION

Carbohydrates are classified according to the hydrolytic products as follows:



Based on number of sugar units present:

- Monosaccharides: Cannot be hydrolyzed further into simpler forms.
- Disaccharides: Yield 2 molecules of same or different monosaccharide units on hydrolysis e.g. sucrose, lactose and maltose.
- Oligosaccharides: Yield 3-10 molecules of monosaccharide units on hydrolysis.
- Polysaccharides: Yield more than 10 molecules of same or different monosaccharide units on hydrolysis.

❖ Homo-& Heteropolysaccharides.

MONOSACCHARIDES

Simplest group of carbohydrates, cannot be further hydrolysed.

Categorization of monosaccharides is based on:

- the Functional Group. (Aldehyde or keto).
- the Number of Carbon atoms.

Molasses (Glucose) | Cherries (Fructose) | Yogurt (Galactose)

MONOSACCHARIDES

No of C-atoms

- Trioses
- Tetroses
- Pentoses
- Hexoses
- Heptoses
- Octoses

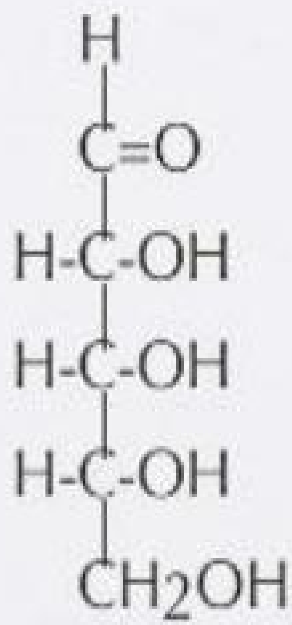
Potentially active carbonyl group

- Aldoses
- Ketoses

"These carbohydrates cannot be hydrolyzed into simpler compounds"

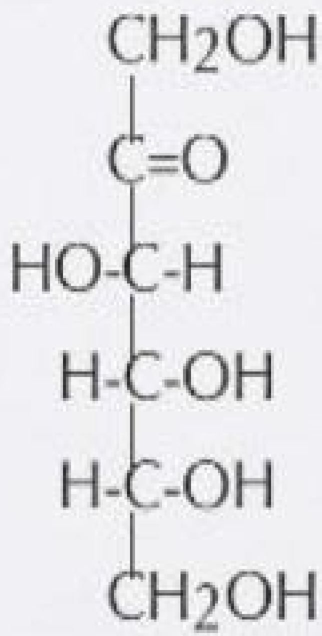
Monosaccharide classifications

Based on location of C=O



Aldose

Aldehyde CHO



Ketose

Ketone C═O

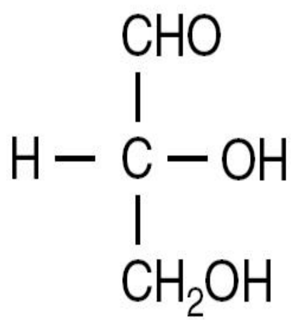
MONOSACCHARIDES BASED ON FUNCTIONAL GROUP

Classification of Monosaccharides

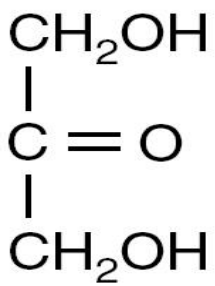
No. of Carbon	Type of sugar	Aldoses	Ketoses
3	TRIOSES	Glyceraldehydes	Dihydroxyacetone
4	TETROSES	Erythrose	Erythrulose
5	PENTOSES	Ribose, Xylose	Ribulose, xylulose
6	HEXOSES	Glucose, Galactose	Fructose
7	HEPTOSES	Glucoheptose	Sedoheptulose

Trioses:

These monosaccharides contain three carbon atoms. The smallest aldose in human body is glyceraldehyde and the smallest ketose in human body is dihydroxyacetone.



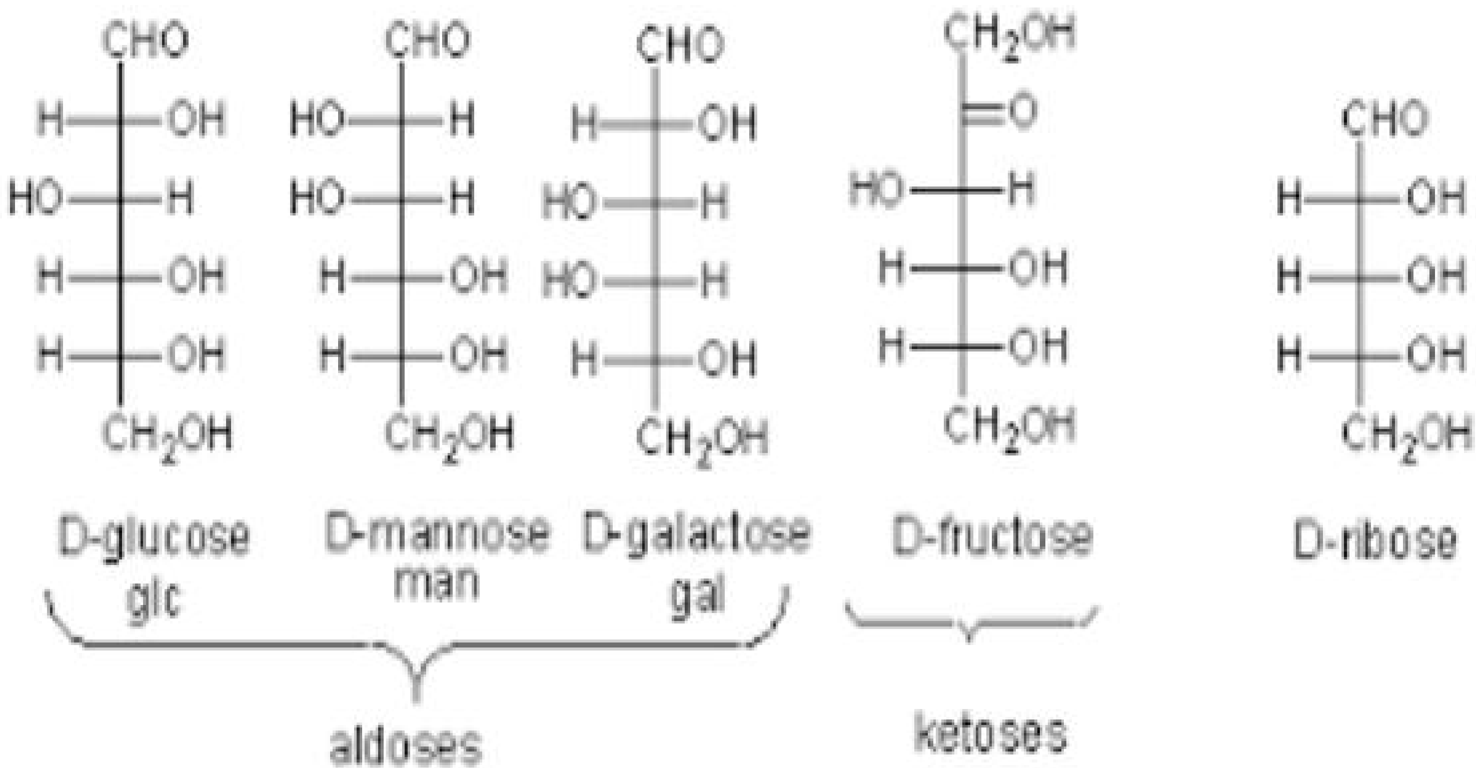
glyceraldehyde



dihydroxyacetone

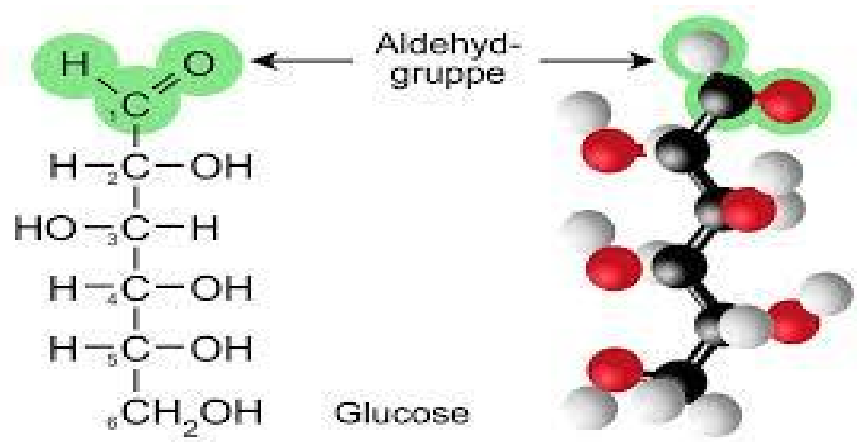
Hexoses:

These monosaccharides contain 6 carbon atoms. The physiologically important hexoses are glucose, galactose, fructose and mannose.



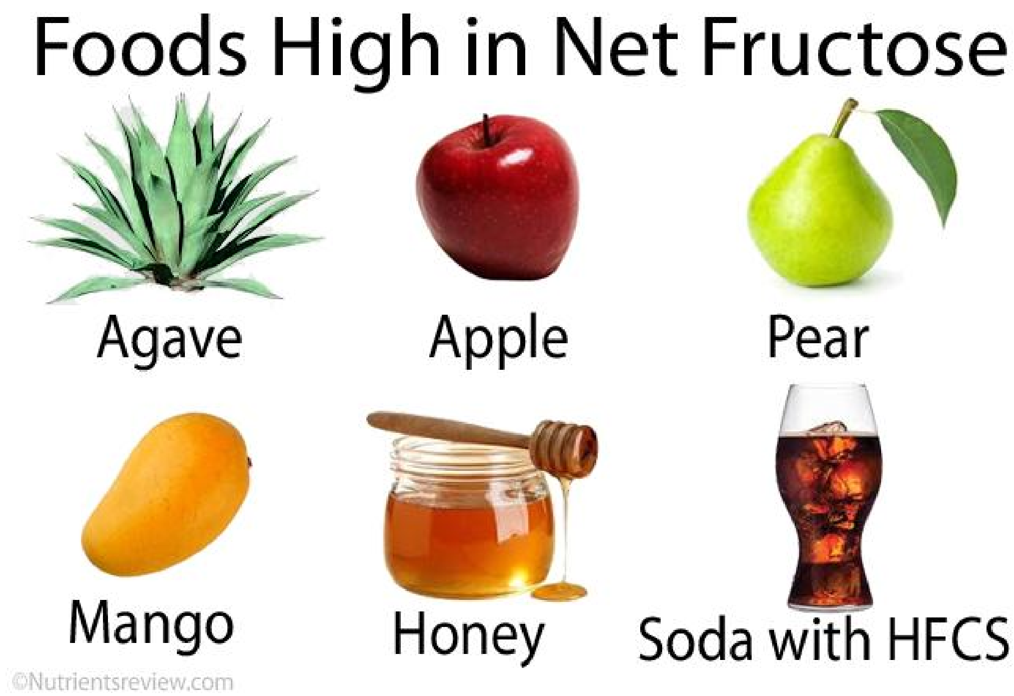
***Glucose*:**

- It is the most important sugar of carbohydrates present in fruit juices and result from the hydrolysis of starch, cane sugar, maltose and lactose.
- Glucose is the sugar carried by the blood and is the principal sugar used by the tissues.
- It is the major source of energy.
- It is present in urine in cases of diabetes mellitus due to raised blood glucose.



Fructose:

- It is present in fruit juices and honey.
- Fructose can be changed to glucose in the liver and intestine, so it can be used in the body.
- Fructose accumulation occurs in hereditary fructose intolerance.



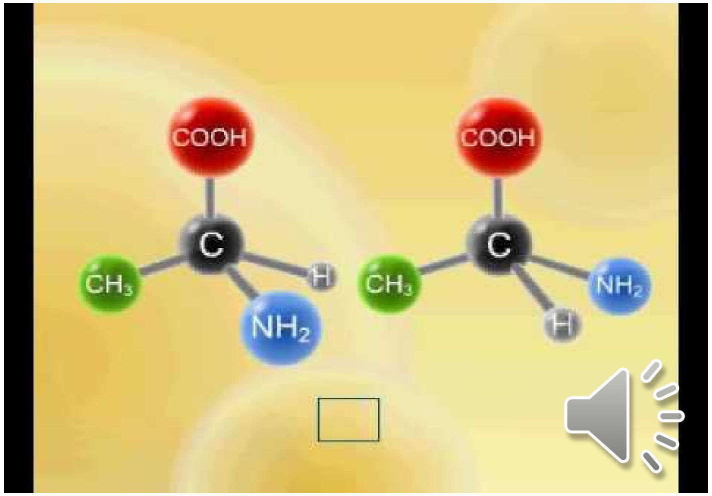
Galactose:

- It is formed during hydrolysis of lactose (milk sugar).
- Galactose can be changed to glucose and metabolized.
- It is synthesized in the mammary gland to make the lactose of milk.
- It is a constituent of glycolipids and glycoproteins.
- Galactosemia and cataract result from failure error of its metabolism.

STEREOISOMERS

❧ Isomers are compounds having the same structural formula but differ in spatial configuration and having different physical properties.

- The formation of isomers is caused by the presence of asymmetric carbon atoms i.e. carbon atoms attached to 4 different atoms.
- The number of possible isomers of a compound depends on the number of asymmetric carbon atoms (n) and is equal to n^2. For example, glucose has 4 asymmetric carbon atoms, so the number of isomers = 16 isomers.
- All monosaccarides have asymmetric carbon atoms except dihydroxyacetone.



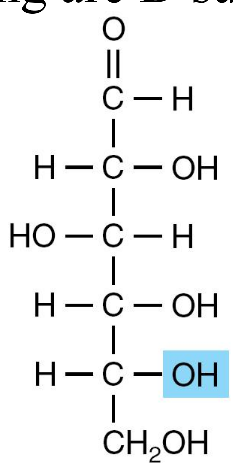
Types of isomerism

(1) D and L isomers:

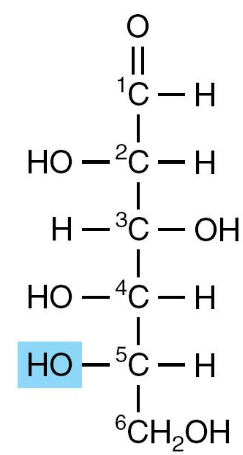
A special type of isomerism in the pairs of structures that are mirror images of each other (Enantiomers).

•This type of isomerism depends upon the orientation of the -H and -OH groups around the carbon atom adjacent to the terminal primary alcohol carbon (C5 in glucose) .

•When the -OH group is on the right, the sugar is a member of the D-series, when it is on the left, It is a member of the L-series. Most of naturally occuring are D sugars.



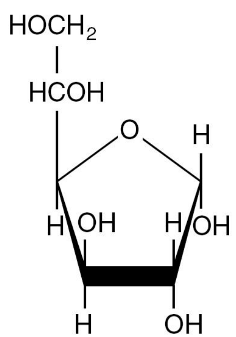
D-Glucose



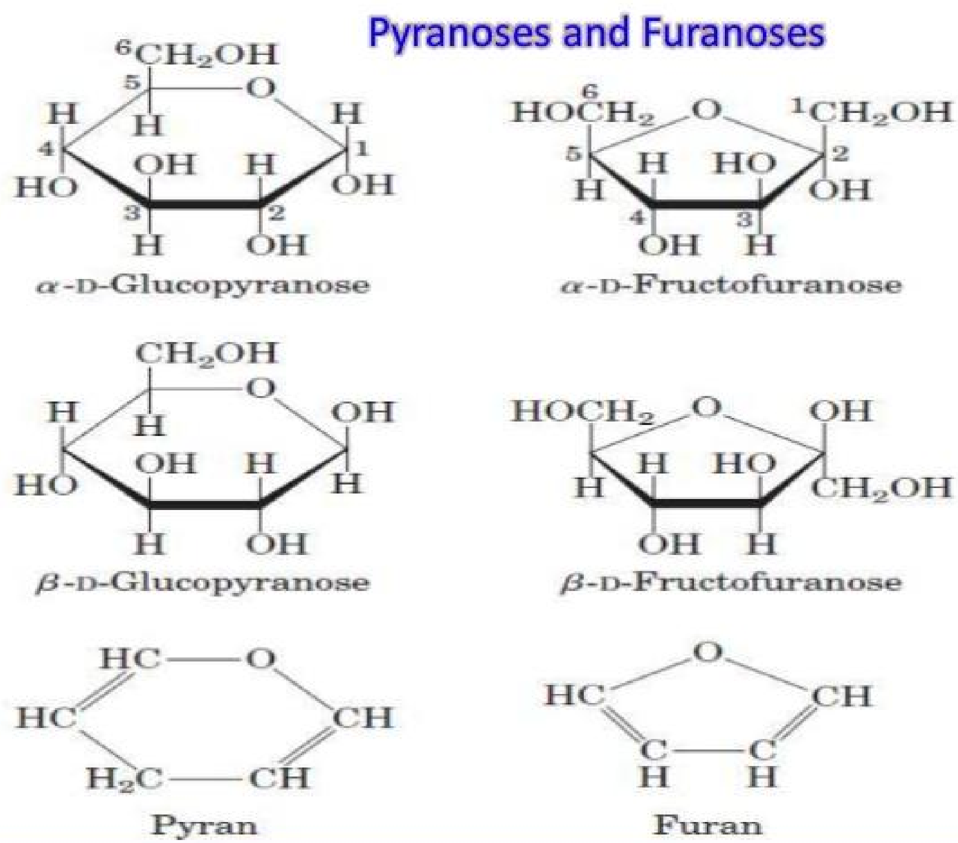
L-Glucose

(2) Pyranose and Furanose ring structure:

- The cyclic structure of glucose is favored and accounts for most of its chemical properties.
- The stable ring structures of monosaccharides are similar to the ring structures of either pyran or furan. 99% of glucose in solution is in the pyranose form.



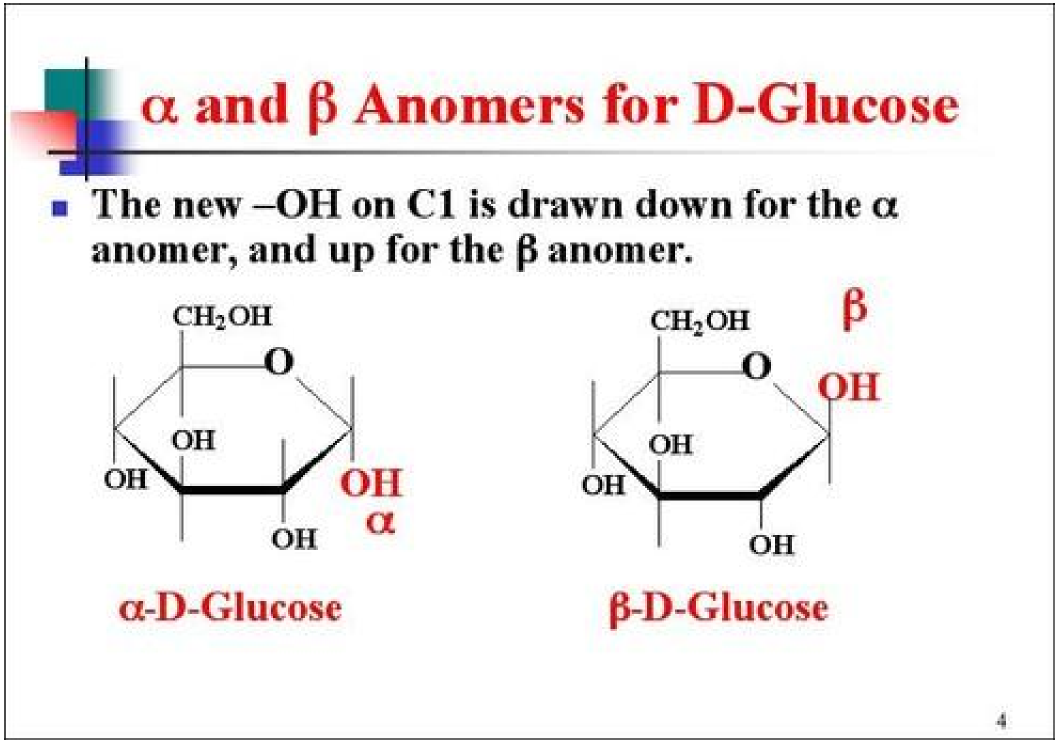
α-D-Glucofuranose



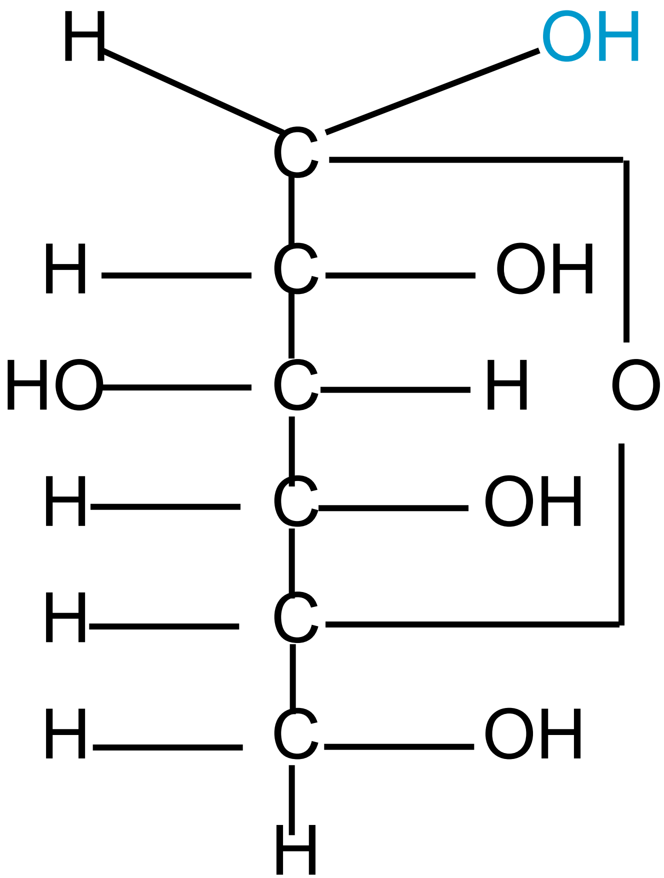
(3) Alpha and beta anomers:

These are isomers differing in the configuration of -OH and -H on C1.

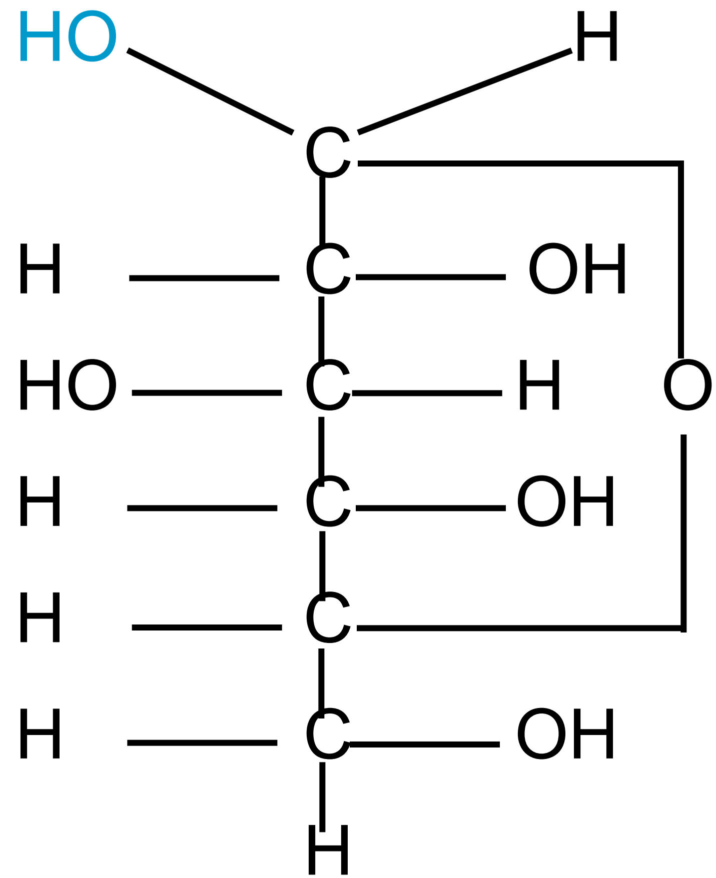
The cyclic structure is retained in solution, but isomerism takes place about the carbonyl or anomeric carbon atoms (C1) to give either alpha-glucopyranose (38%) or beta-glucopyranose (62%).



Alpha and Beta anomers of glucose



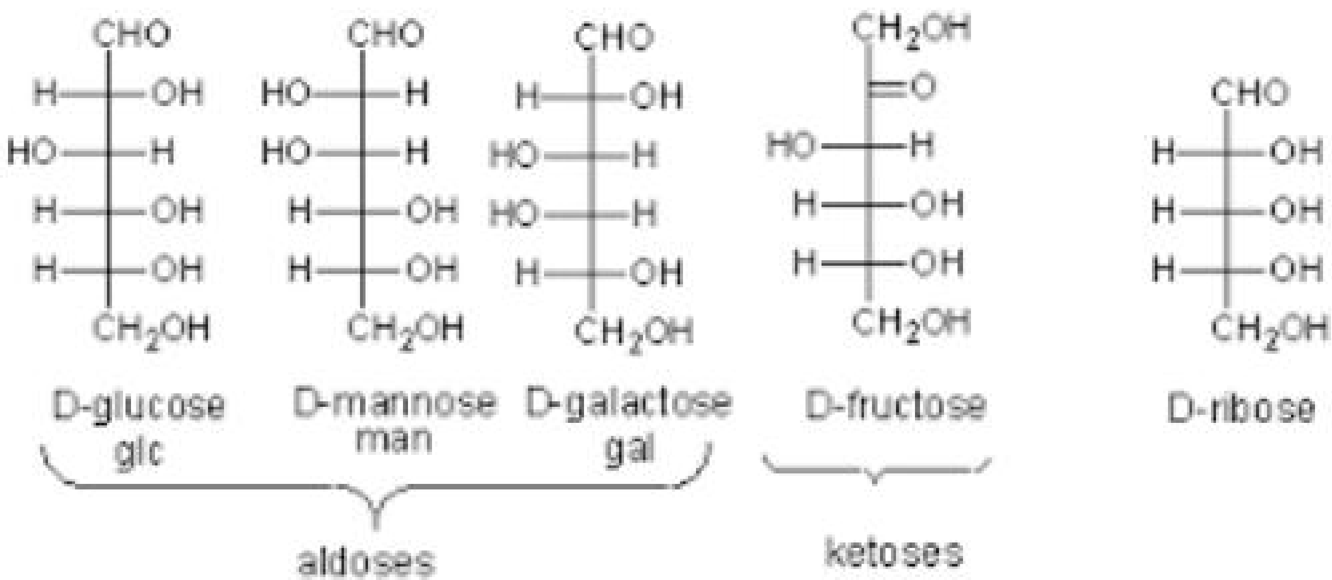
α-D-GLUCOPYRANOSE



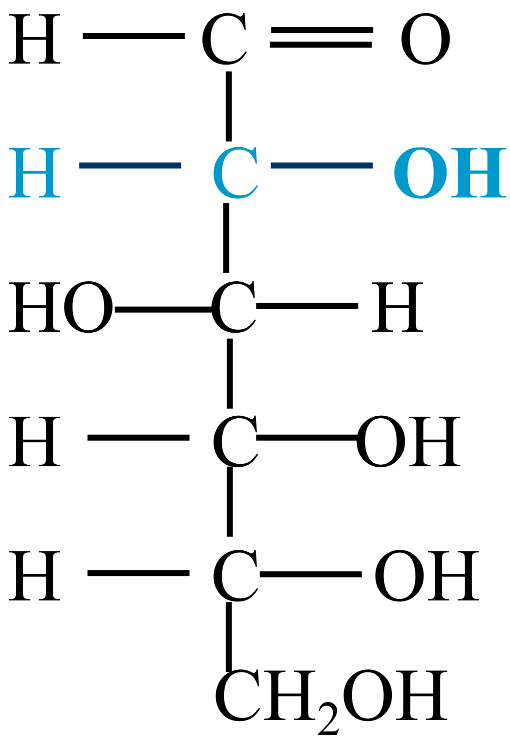
β-D-GLUCOPYRANOSE

(4) Epimers:

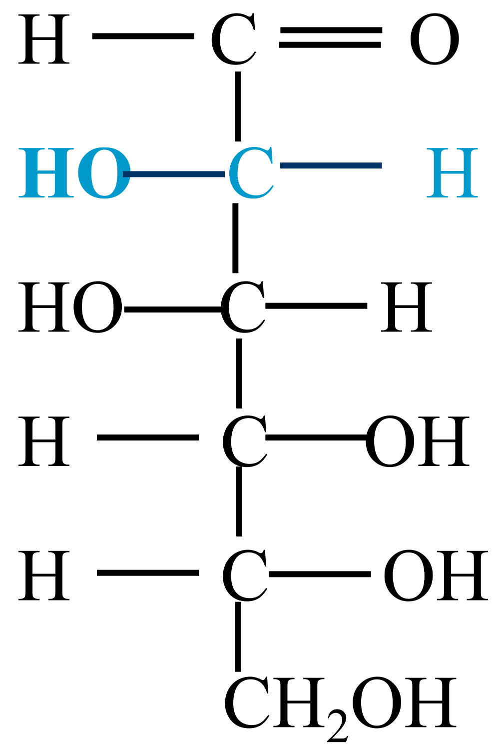
- These are isomers differing in the configuration of the -OH and -H on carbon atoms 2, 3 and 4 of glucose.
- The most important epimers of glucose are mannose and galactose. Mannose is formed by epimerization at carbon 2 whereas, galactose by epimerization at carbon 4.



CARBON-2 EPIMERS

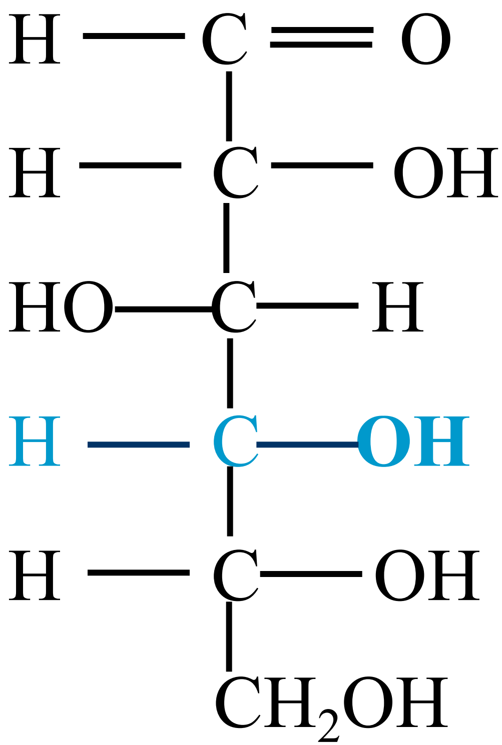


D-GLUCOSE

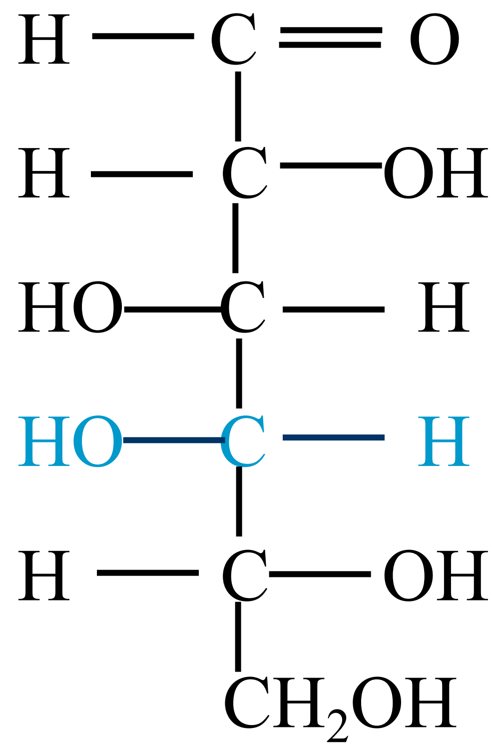


D-MANNOSE

CARBON-4 EPIMERS



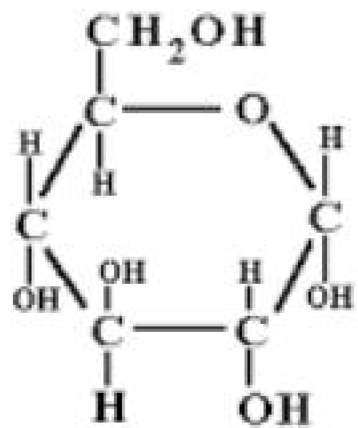
D-GLUCOSE



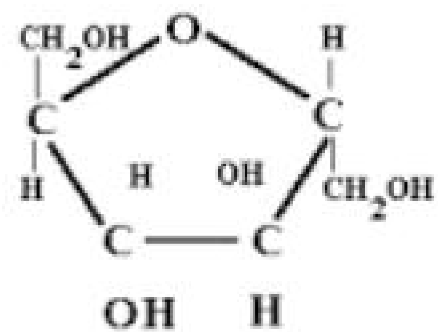
D-GALACTOSE

(5) Aldoses and Ketoses:

- Depending on the presence of aldehyde or keto group in the sugar.
- Fructose contains ketone group in position 2, whereas there is an aldehyde group in position 1 of glucose.



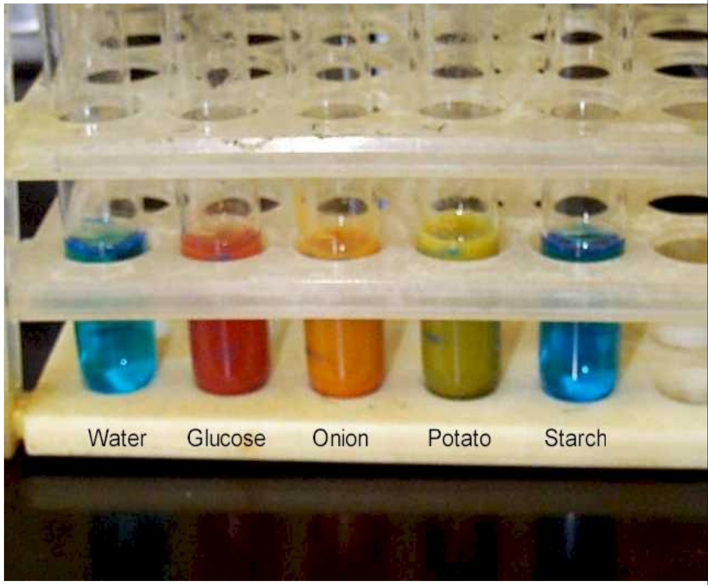
Glucose



Fructose

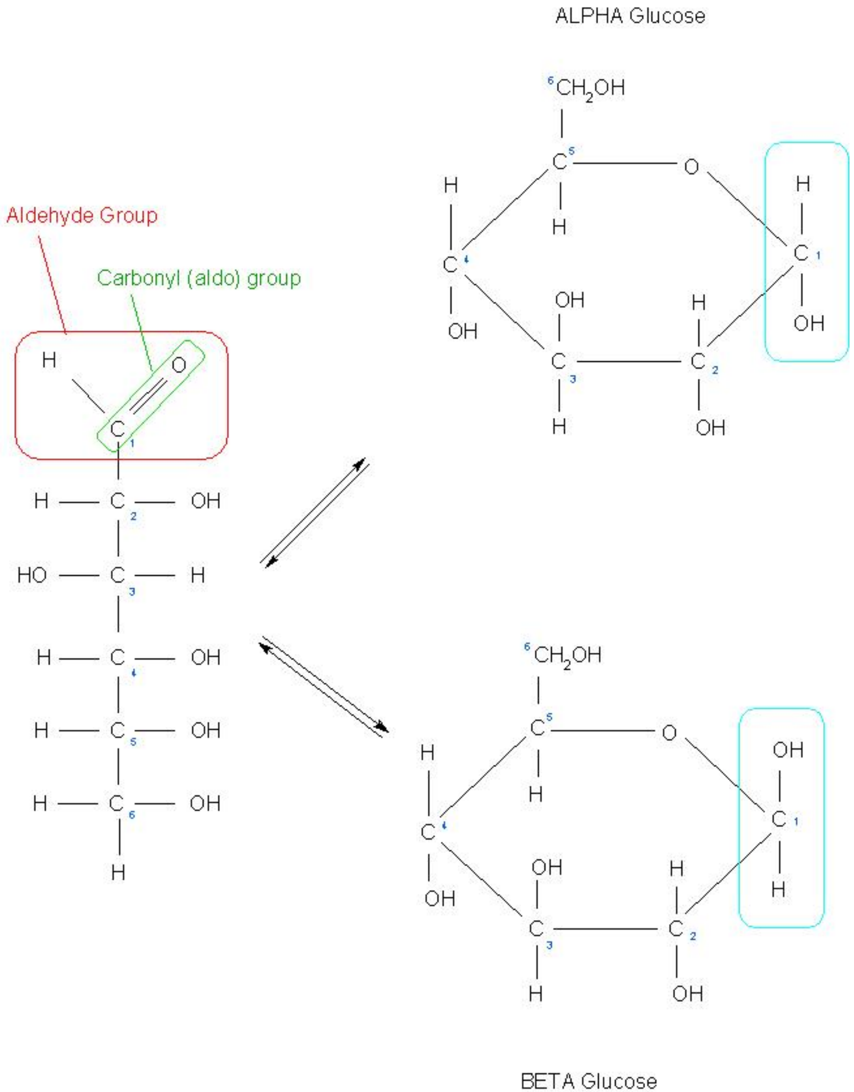
Benedict’s Test

- Benedict’s reagent undergoes a complex colour change when it is reduced
- The intensity of the colour change is proportional to the concentration of reducing sugar present
- The colour change sequence is:
  - Blue…
  - green…
  - yellow…
  - orange…
  - brick red



The carbonyl group - monosaccharides

- The carbonyl group is “free” in the straight chain form.
- But not free in the ring form.
- BUT remember – the ring form and the straight chain form are interchangeable.
- So **all** monosaccharides are reducing sugars.
- All monosaccharides reduce Benedict’s reagent.



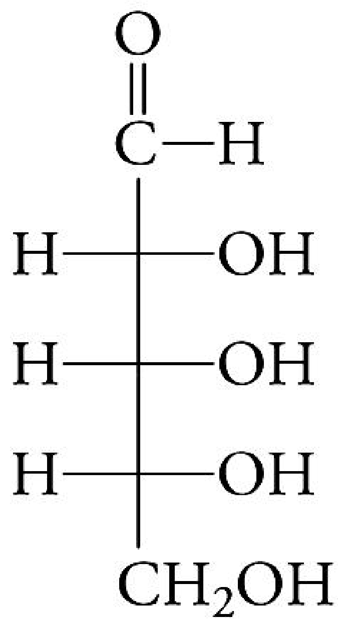
Optical activity

- Optical activity is the ability of a compound to rotate plane polarized light either to the left [-] (levorotatory) or to the right [+] (dextrorotatory).
- The plane- polarized light (PPL) can be obtained by passing a beam of ordinary light through a NiKol's prism ($CaCO_3$).
- The naturally occuring form of glucose is D (+) isomer.
- The compounds having asymmetric carbon atoms can rotate the beam of plane polarized light and are said to be optically active.
- DL-mixture is a mixture containing equal amounts of D and L isomers, the resulting solution has no optical activity since the activities of each isomer cancel one another. Such structure is called racemic or DL-mixture.
- All monosaccharides are optically active except dihydroxyacetone because they contain asymmetric carbon atoms.

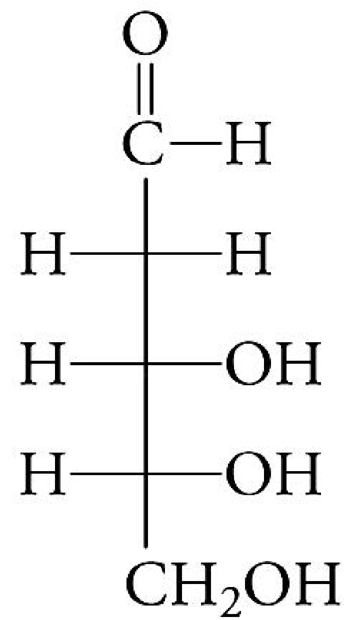
Monosaccharide Derivatives

Deoxy sugars:

- are sugars in which a hydrogen atom replaces one or more of the -OH groups in a monosaccharide.
- D-ribose and its derivative D-2-deoxyribose (deoxy = minus one oxygen atom) are found in various coenzymes and in DNA.



D-Ribose



D-2-Deoxyribose

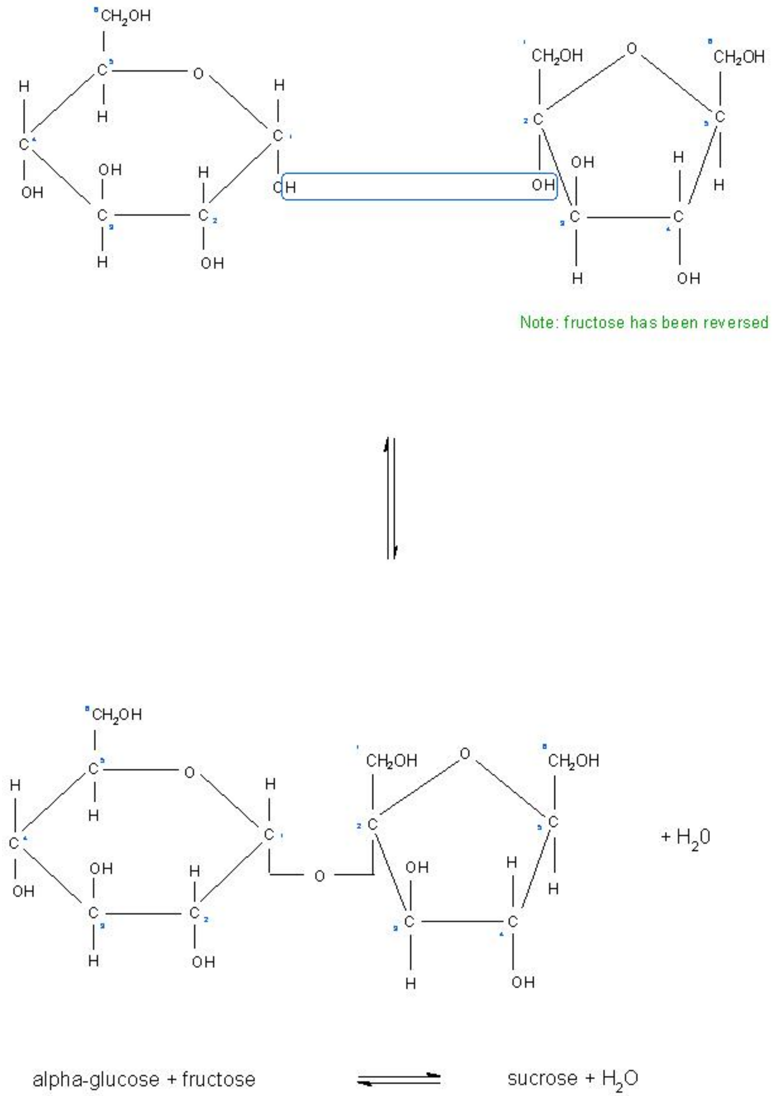
DISACCHARIDES

- These are sugars formed of 2 monosaccharide units linked by a glycosidic linkage.
- The physiologically important disaccharides are maltose, sucrose and lactose.

- **Reducing:** Maltose, Lactose –with free aldehyde or keto group.
- **Non-reducing:** Sucrose, Trehalose –no free aldehyde or keto group.

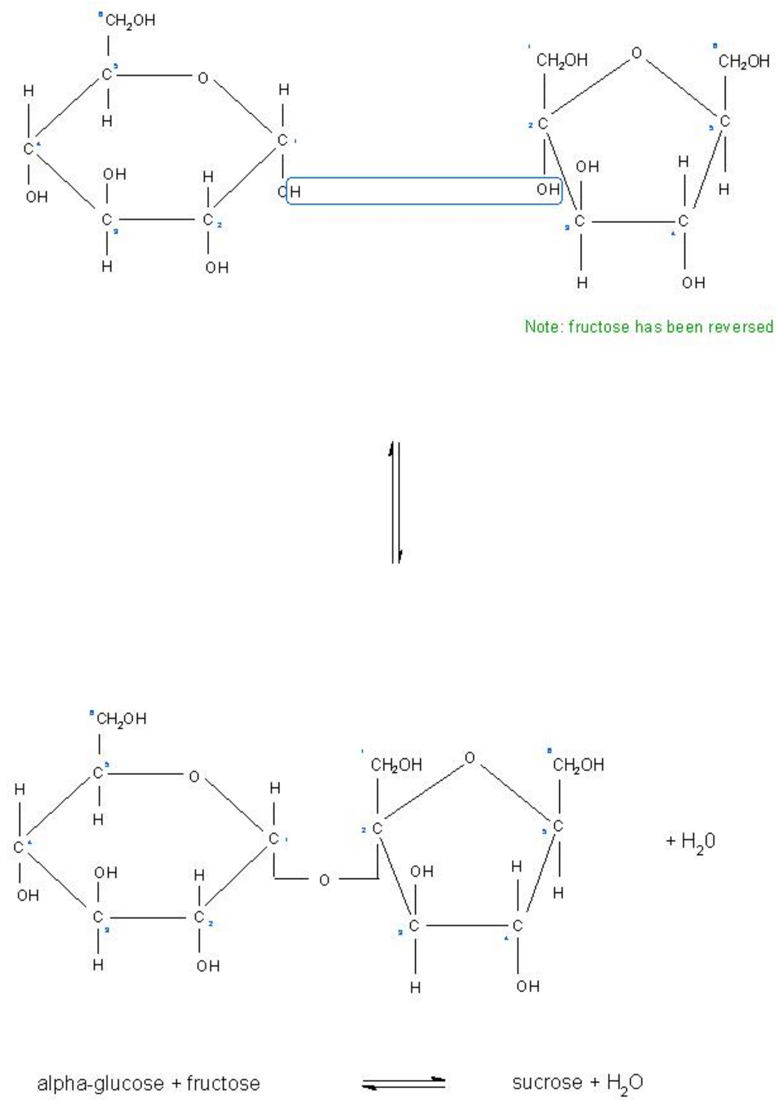
The carbonyl group – disaccharides - sucrose

- In some disaccharides e.g. sucrose **both** of the carbonyl groups are involved in the glycosidic bond.
- So there are **no** free carbonyl groups.
- Such sugars are called **non-reducing sugars.**
- They **do NOT** reduce Benedict's reagent.



The carbonyl group – disaccharides - sucrose

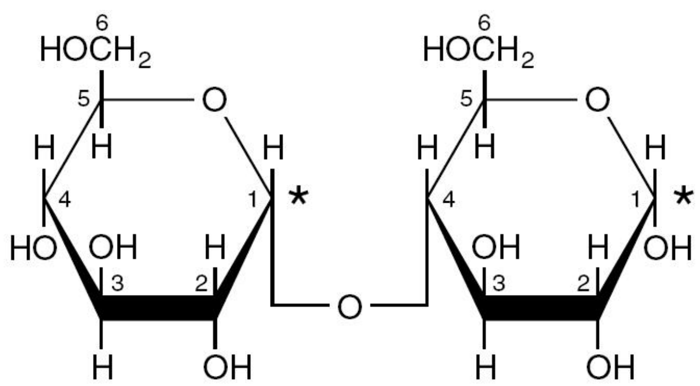
- The subunits of sucrose (glucose and fructose) are reducing sugars.
- If sucrose is hydrolysed the subunit can then act as reducing sugars.
- This is done in the lab by **acid hydrolysis.**
- After acid hydrolysis sucrose **will** reduce Benedict’s reagent.



(1) Maltose:

•It is composed of 2 alpha D- glucose units united by alpha 1-4 glycosidic linkage.

•Maltose is found in germinating cereals and malt. It is produced during hydrolysis of starch by amylase.

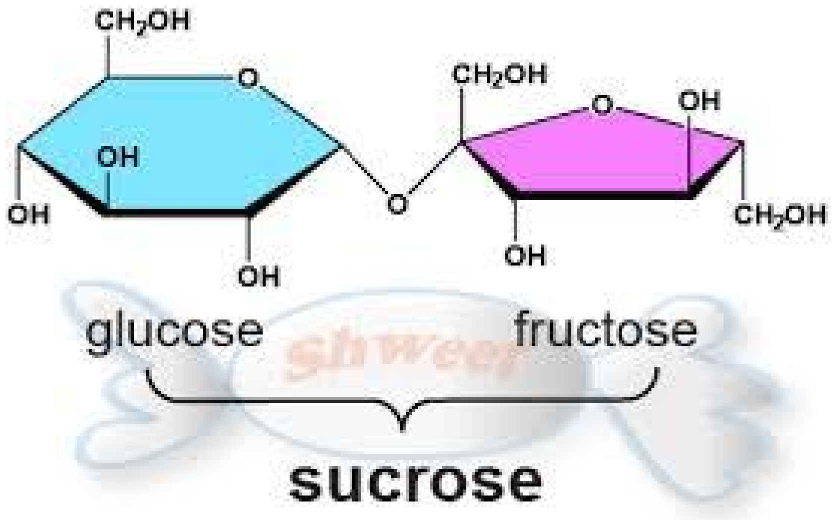


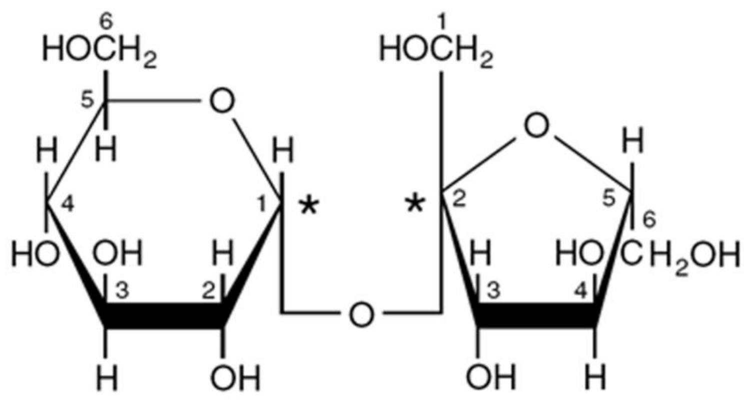
α-D-Glucose **α-D-Glucose**

(2) Sucrose:

- It is called cane sugar or table sugar.
- Sucrose is formed by condensation of a molecule of alpha-D glucose with a molecule of beta-D fructose united by alpha 1-2 glycosidic linkage.
- It is dextrorotatory. On hydrolysis by the invertase (sucrase) enzyme, it yields a levorotatory mixture called invert sugar because the strong levorotatory of fructose inverts the previous dextrorotatory action of sucrose.
- It is non-reducing sugar because the glycosidic linkage involves the carbonyl groups of fructose and glucose.

Sucrose

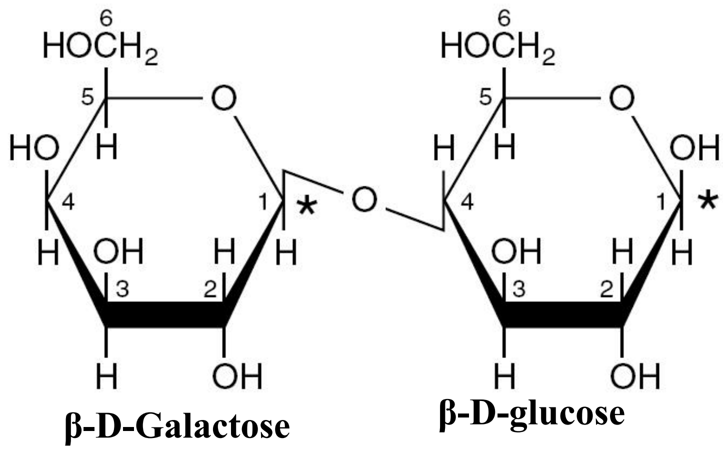




α-D-Glucose β-D-fructose

(3) Lactose (milk sugar):

- It is formed by condensation of a molecule of beta-D galactose with a molecule of beta D- glucose united by beta-1-4-galactosidic linkage.
- It is found in milk and may occur in urine during pregnancy.
- Lactose is hydrolyzed by the enzyme lactase. In lactase deficiency, malabsorption of lactose leads to diarrhea and flatulence.
- Least sweet sugar – 16% of sucrose.

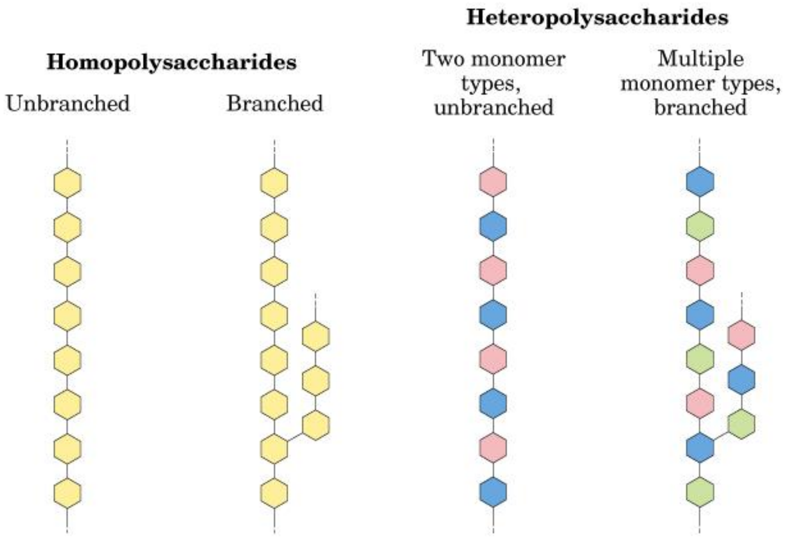


POLYSACCHARIDES

- Polysaccharides are formed by condensation of more than 10 monosaccharide units.
- They form colloidal solutions in water because they have high molecular weights.
- They are classified into 2 groups:

  (A) Simple or homopolysaccharides.

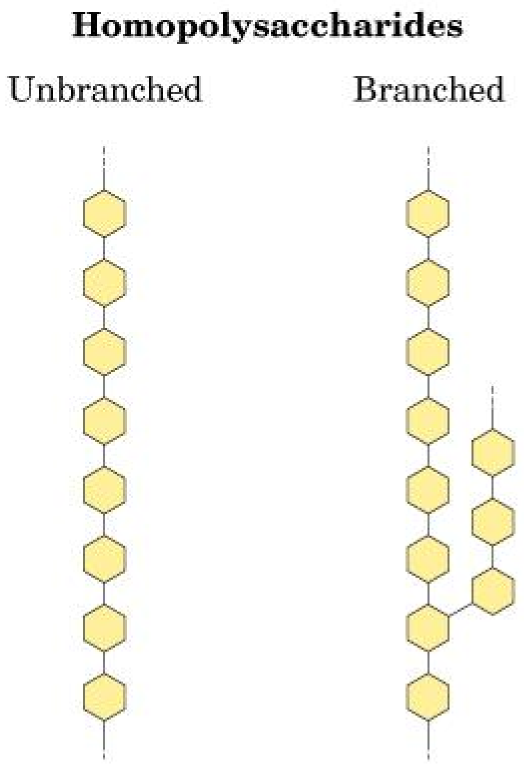
  (B) Mixed or heteropolysaccharides.



(A) Homopolysaccharides:

- These are formed of only one type of sugar units. Homopolysaccharides include the following physiologically important carbohydrates:

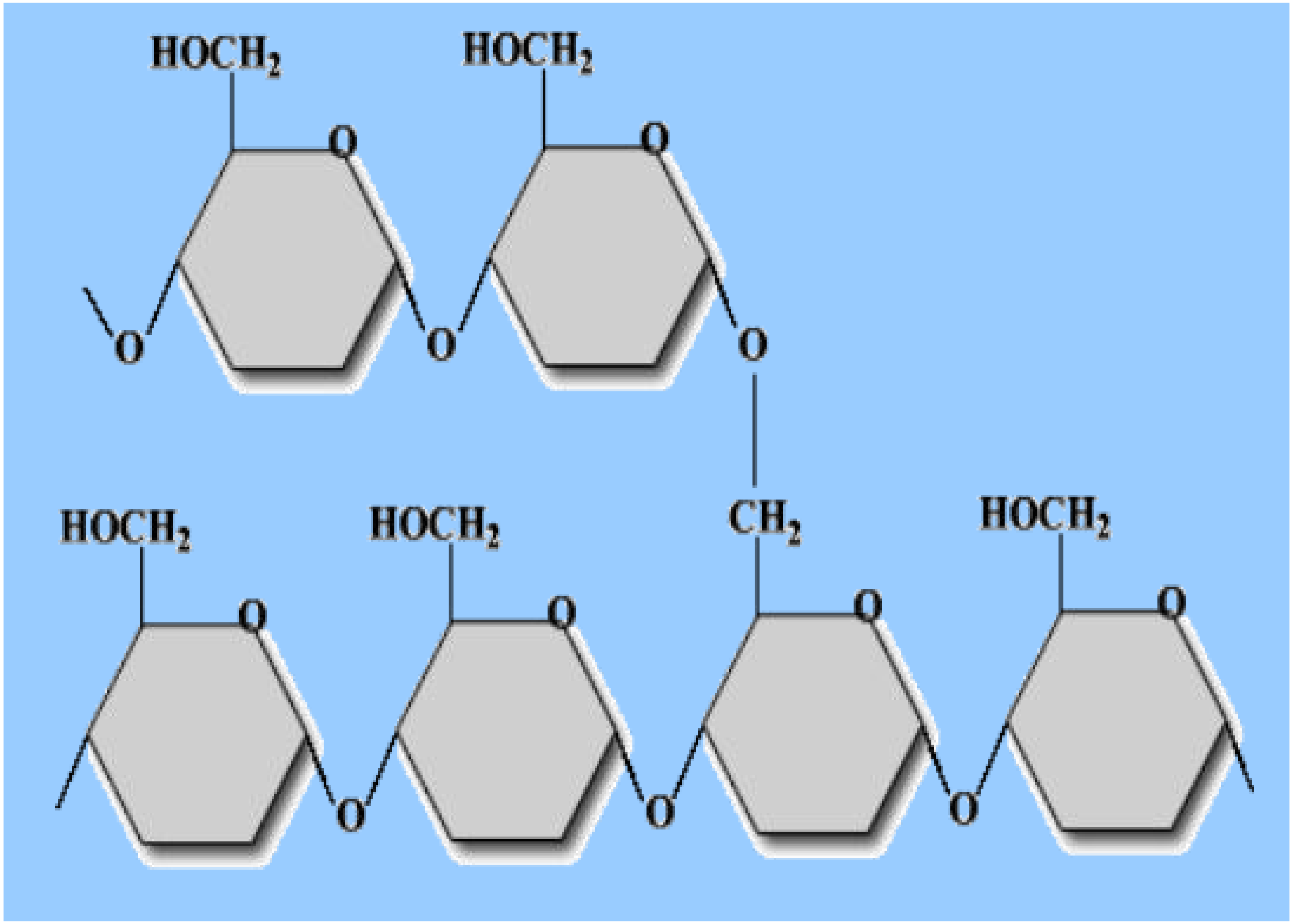
- **Starch**
- **Glycogen**
- **Cellulose**
- **Inulin**
- **Dextrans**
- **Chitin**
- **Agar**



Starch

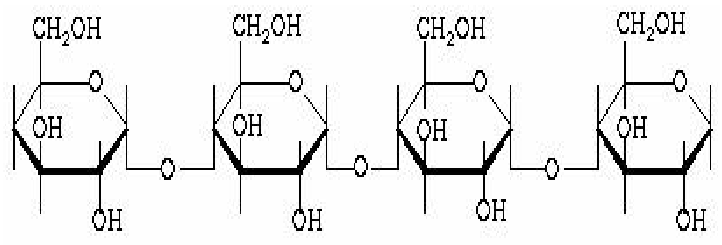
- Main storage form of glucose in plants
- Polysaccharide units
  - Amylose (20—28%)
  - Amylopectin 72—80%)
- Polymer of α-D-glucose
- α-1 $\longrightarrow$ 4 glycosidic linkage
- At branching points α-1 $\longrightarrow$ 6 linkage
- No free aldehyde group
- Found in wheat, rice, corn, potatoes

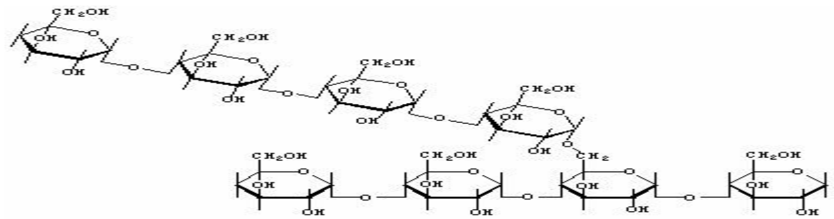
Starch



Characteristics of Amylose and Amylopectin

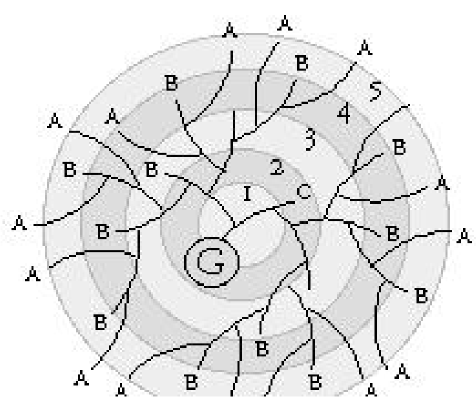
Characteristic	Amylose	Amylopectin
Form	Linear	Branch
Linkage	α-1,4 (some α-1,6)	α-1,4; α-1,6
Polymer units	200-2,000	Up to 2,000,000
Molecular weight	Generally <0.5 million	50-500 million
Gel formation	Firm	Non-gelling to soft

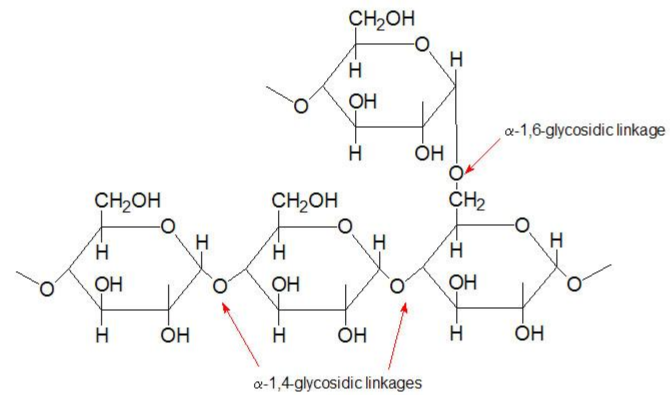




Glycogen

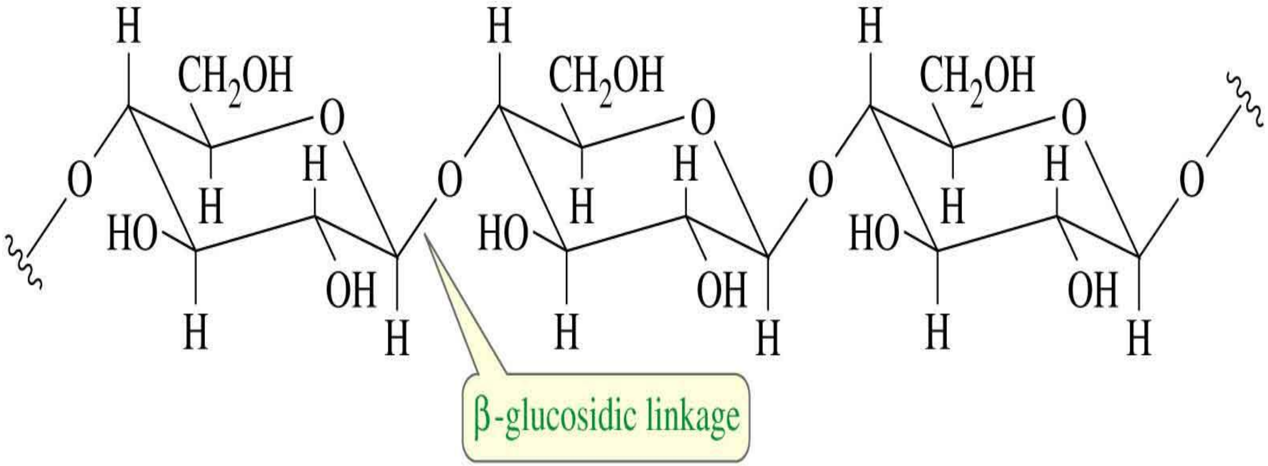
- Glycogen is the chief storage form of carbohydrates in animals known as animal starch.
- Present in the liver and muscle.
- Both α-1 ⟶ 4 and α-1 ⟶ 6 linkages are found.
- More branched structure than starch.
- Gives red color with iodine.





Cellulose

- Cellulose is the chief constituent of the cell wall of plants. It is insoluble and consists of beta-D glucose units linked by beta (1-4) glycosidic bonds. It gives no color with iodine.
- Cellulose can not be digested by many mammals, including humans because of the absence of cellulase enzyme that attacks the beta-linkage, It increases the bulk of stool which increases the intestinal movement, preventing constipation.



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Inulin

- Inulin is found in roots.
- It is fructosan. Can hydrolyzed to fructose.
- bonds. It gives no color with iodine.
- It is used to detect GFR

Dextrins

- It is result from hydrolysis of starch
- It is used to detect GFR

chitin

It is present in exoskeleton of insect

Agar

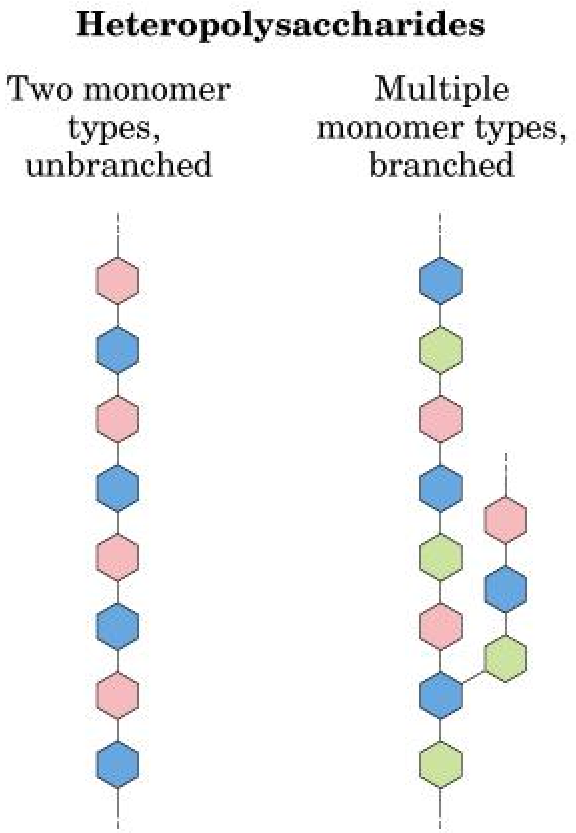
Is galactosan found in algae . It is used as solid medium for bacterial growth

(B) Heteropolysaccharides:

- These are polysaccharides formed of more than one type of simple sugar units. These include: **glycosaminoglycans and glycoproteins**.

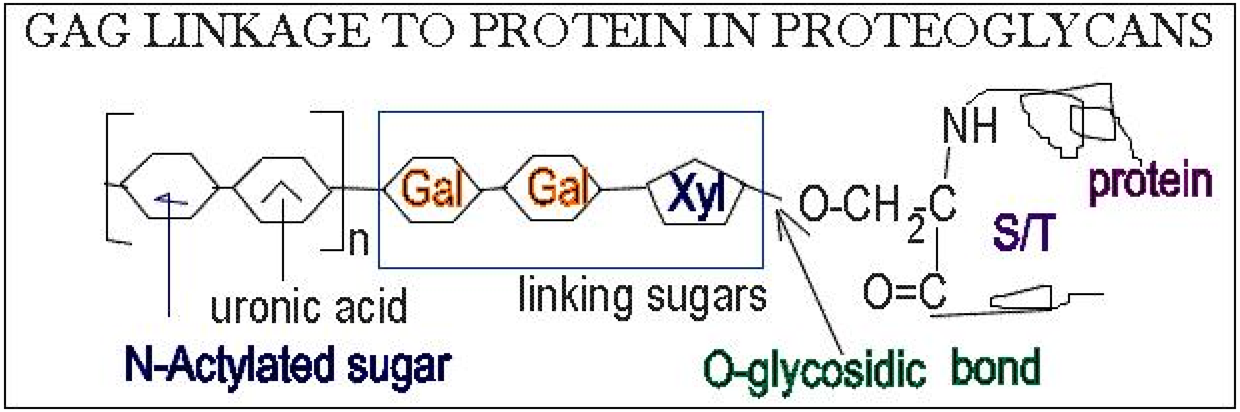
Mucopolysaccharides:

- **Hyaluronic acid**
- **Heparin**
- **Chondroitin sulphate**
- **Keratan sulphate**
- **Dermatan sulphate**



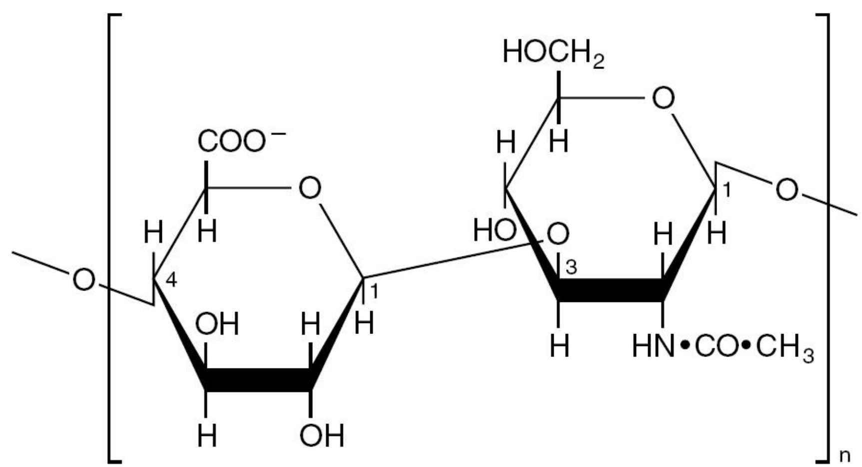
(A) Glycosaminoglycans (Mucopolysaccharides):

- Glycosaminoglycans consist of chains of repeating disaccharides units complex carbohydrates which are characterized by their content of amino sugars and uronic acids.
- When these chains are linked to a protein molecule, the compound is known as proteoglycan.
- Glycosaminoglycans are associated with bone, elastin and collagen which are the structural elements of the tissues.



(1) Hyaluronic acid:

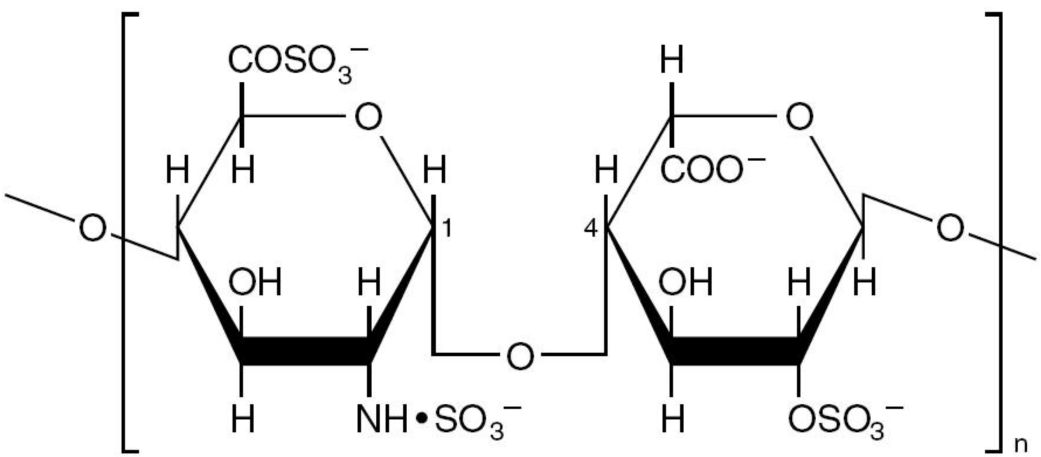
- Hyaluronic acid is formed of alternate units of N-acetyl glucosamine and glucuronic acid, united by beta 1→3 glucosidic linkage.
- It is present in bacteria and is widely distributed among various animals and tissues including synovial fluid, the vitreous humour of the eye and loose connective tissues.
- The hyaluronic acid in connective tissue can be hydrolyzed by an enzyme called hyaluronidase or spreading factor. This enzyme is secreted by certain bacteria and is present in the acrosomal cap of the sperms helping the process of fertili



β-Glucuronic acid   N-Acetylglucosamine

(2) Heparin:

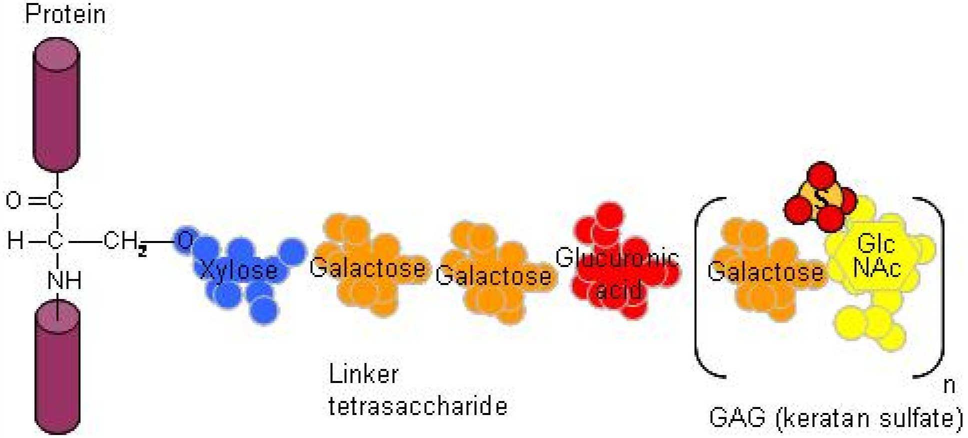
- Heparin consists of repeating disaccharide units containing glucosamine (GLcN) and either glucuronic (10%) or iduronic acids (90%).Most of the amino groups of GLcN are N-sulfated.
- The protein molecule of heparin proteoglycan consists exclusively of serine and glycine residues.
- Heparin is found in the granules of mast cells and also in liver, lung and skin. It has anticoagulant properties and activates plasma lipoprotein lipase enzyme.



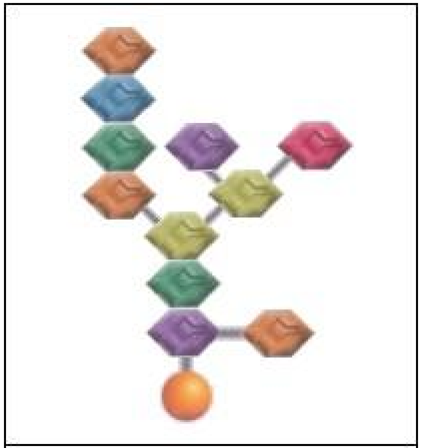
Sulfated glucosamine **Sulfated iduronic acid**

(B) Glycoproteins:

- Glycoproteins are proteins containing carbohydrates in varying amounts (2-15 units which is usually branched).
- Carbohydrate content ≤ 10%.



Proteoglycan



Glycoform attached to a protein (Glycoprotein)

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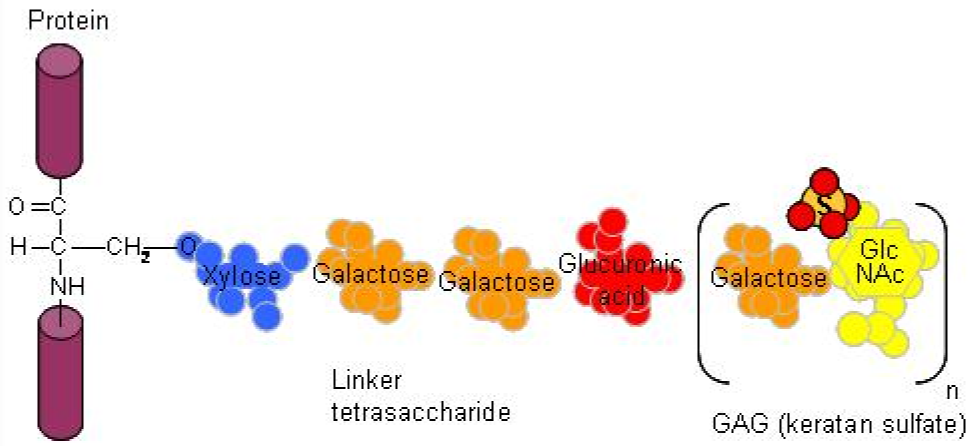
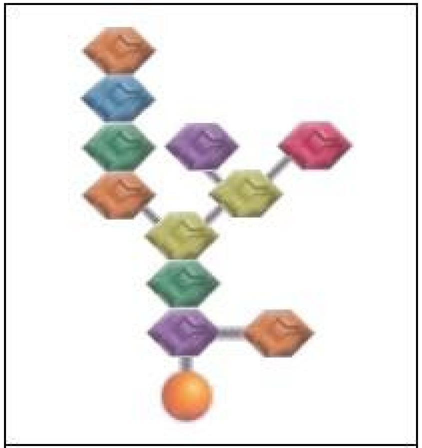


Proteoglycan



Glycoform attached to a protein (Glycoprotein)

Proteoglycans

- Mainly carbohydrates 85% with 15 percent protein.
- Composed of REPEATING units – heteropolysaccharide
- Carbohydrates are linear with more than 50 sugar units

Glycoproteins

- Mainly protein with minimal carbohydrates
- Composed of varying monosaccharide units.
- Carbohydrates are branched with 3-10 sugar units.

THANK YOU!